

Bond MV, Price, and YTM Explanation

Formula 1 – Compound Interest (for investments, not coupon bonds): $MV = P \times (1 + r)^n$

Where: - MV = Maturity Value (total amount at the end) - P = Principal (starting amount) - r = interest rate per compounding period - n = total number of compounding periods

Example: P = 500,000 Annual rate = 11% Semiannual compounding $\rightarrow r = 0.11 / 2 = 0.055$ Term = 10 years $\rightarrow n = 20$

Step-by-step: $MV = 500,000 \times (1 + 0.055)^{20}$
 $1 + 0.055 = 1.055$
 $1.055^{20} \approx 2.918$
 $MV \approx 500,000 \times 2.918 = 1,459,000$

Formula 2 – Price of a Bond with YTM: $\text{Price} = \sum [\text{Coupon Payment} / (1 + \text{YTM_per_period})^t] + [\text{Face Value} / (1 + \text{YTM_per_period})^n]$

Where: - Coupon Payment = Face Value $\times$ (Coupon Rate / periods per year) - YTM_per_period = annual YTM / periods per year - t = each period until maturity - n = total number of periods

Example (semiannual, 10 years): Face Value = 500,000 Coupon Rate = 11% $\rightarrow 27,500$ per half-year
YTM = 8.25% annual $\rightarrow 4.125\%$ per half-year n = 20 periods

Price today: $\text{Price} = \sum [27,500 / (1 + 0.04125)^t] + [500,000 / (1 + 0.04125)^{20}]$

Maturity Value (MV) for a coupon bond: $MV = \text{Face Value} + \text{Final Coupon Payment} = 500,000 + 27,500 = 527,500$